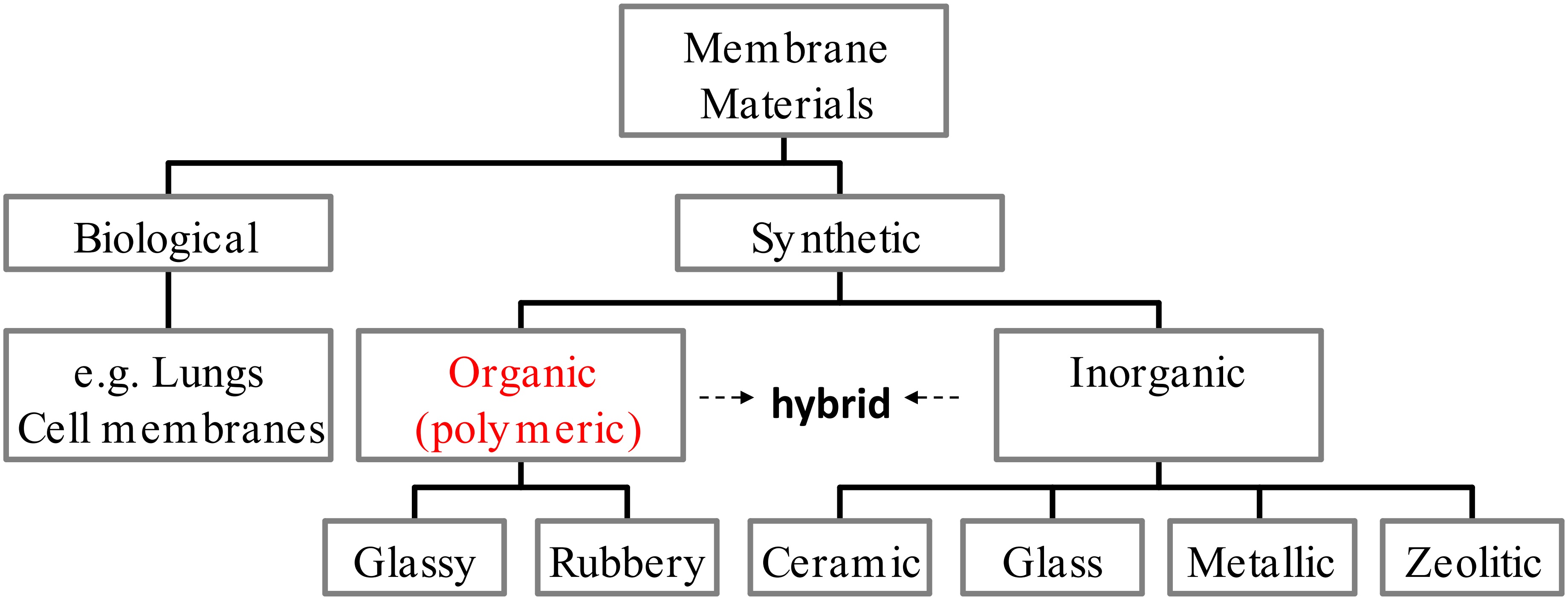
Introduction to Polymer Materials

Lecture 12: Membranes

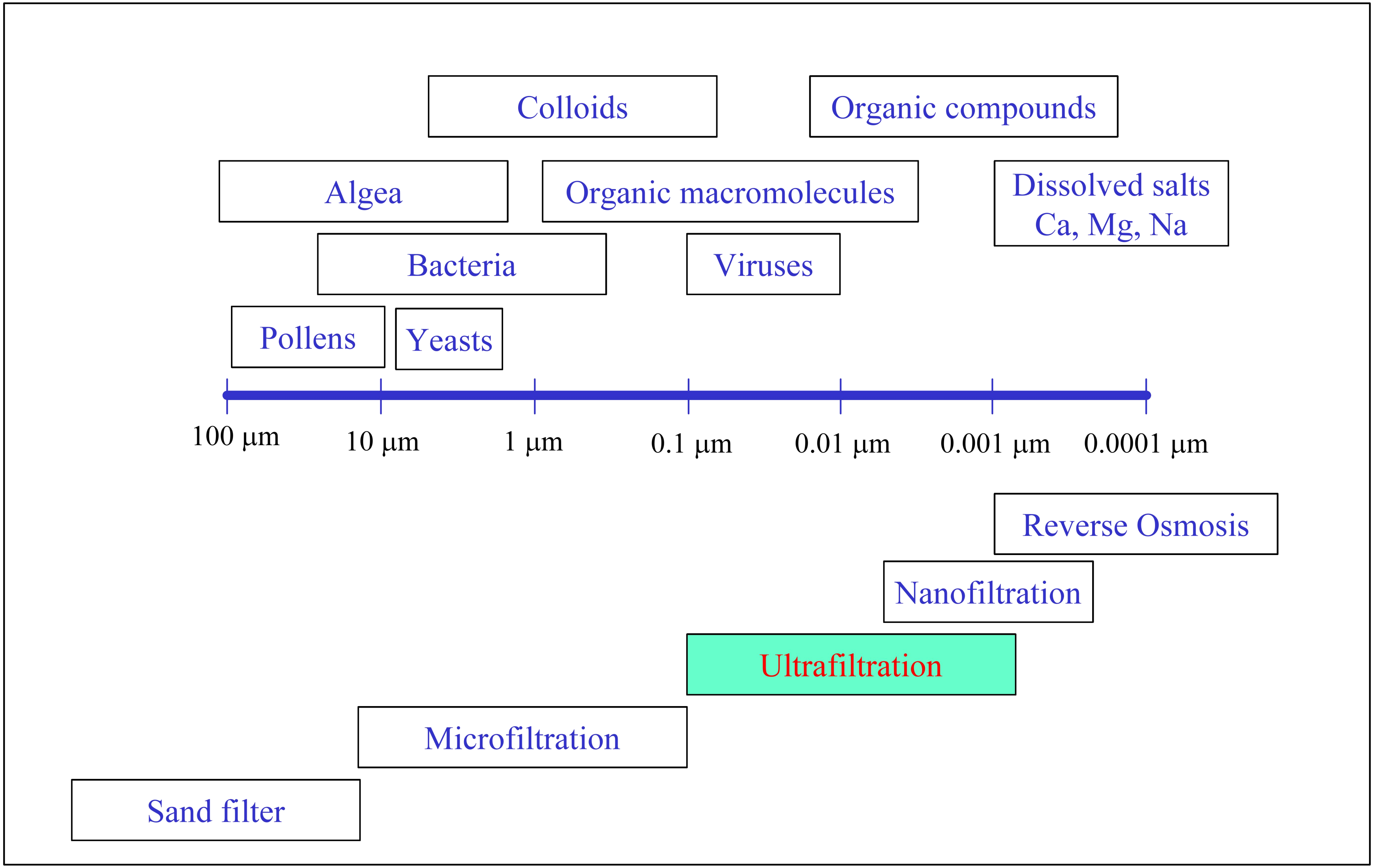
SUN Dazhi

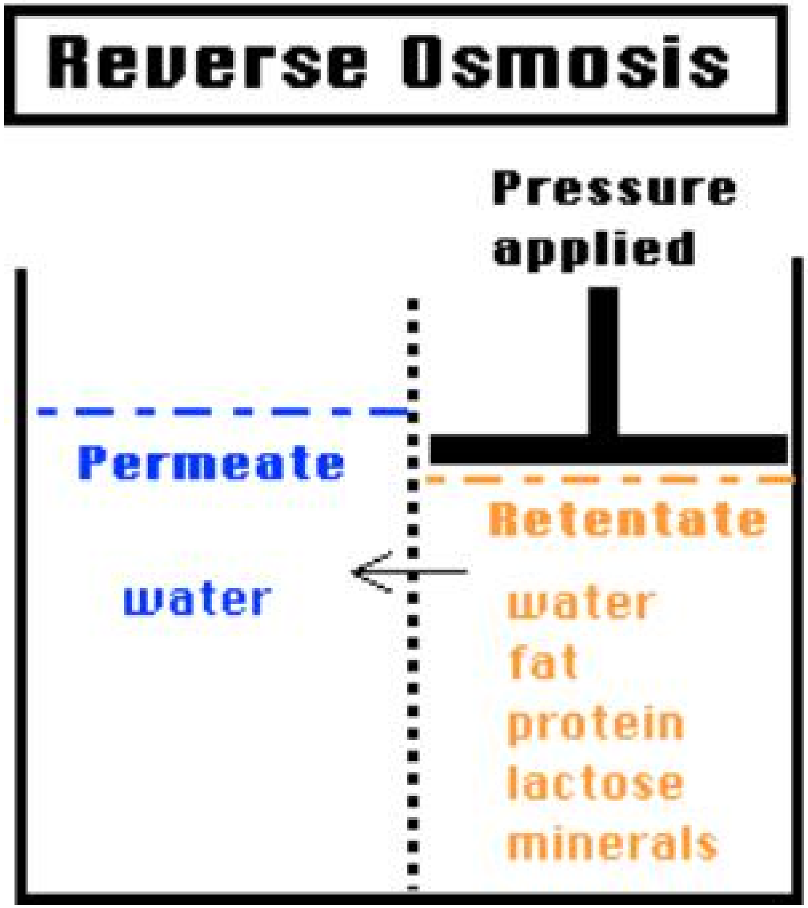
Membranes - Material Selection

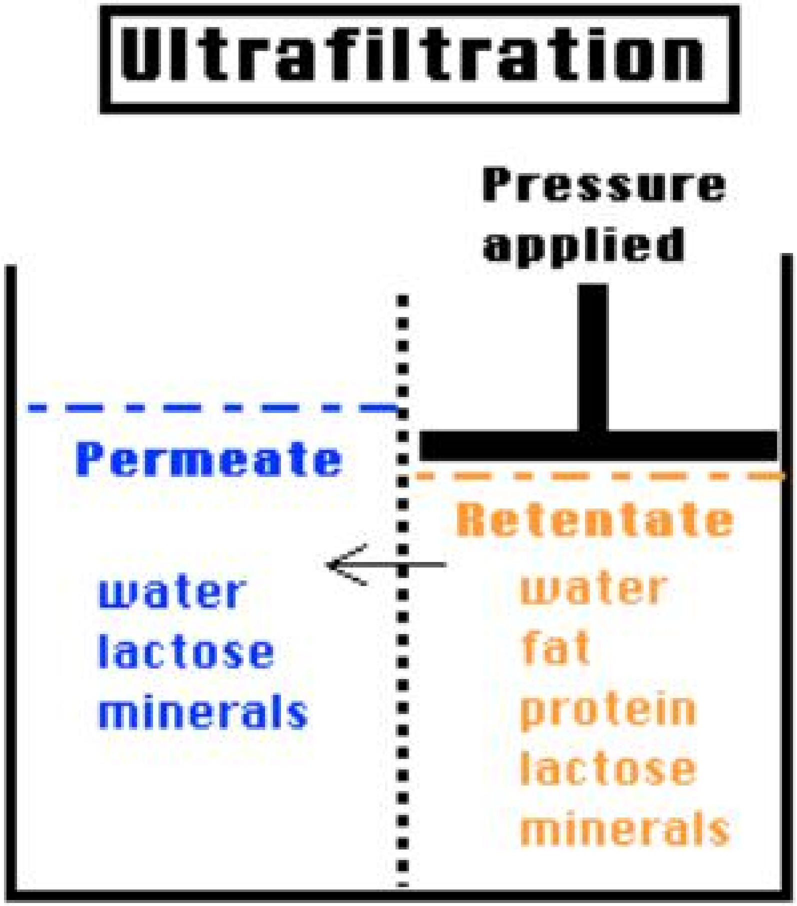
Polymers: most common
Inorganic: more stable



Membranes - How Big



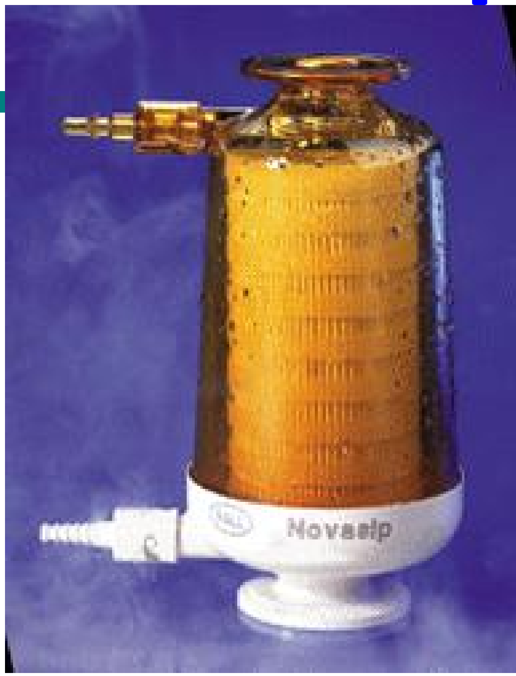

Reverse Osmosis
Pressure applied
Permeate
water
Retentate
water
fat
protein
lactose
minerals

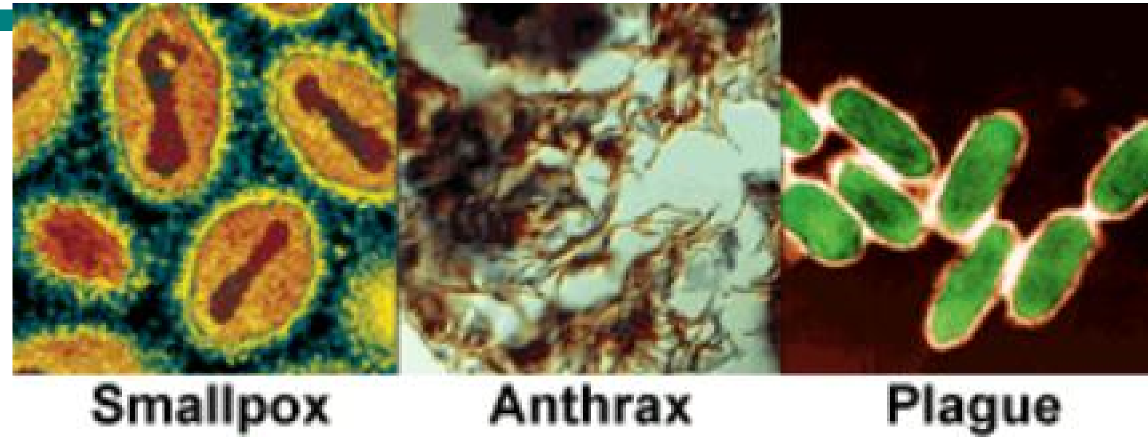

Ultrafiltration
Pressure applied
Permeate
water
lactose
minerals
Retentate
water
fat
protein
lactose
minerals

Size-Dependent

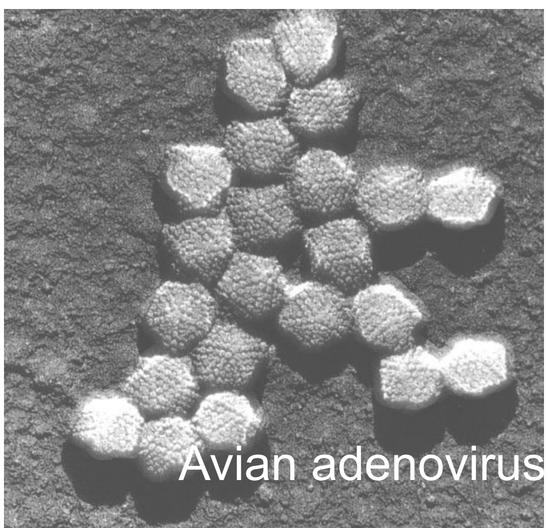
- **Microfiltration (MF)**
  - Micron sized pores
  - Mainly used for particle-fluid separation
  - TMP: 1 to 50 psig
- **Ultrafiltration (UF)**
  - Pores: 10 – 1000 angstroms
  - Used for: Concentration, desalting, clarification and fractionation
  - TMP: 10 – 100 psig
- **Nanofiltration (NF)**
  - TMP: 40 – 200 psig
- **Reverse osmosis (RO)**
  - TMP: 200 – 300 psig
- **Dialysis**
  - Concentration gradient driven
  - Selectivity based indirectly on size

Pathogen Filters



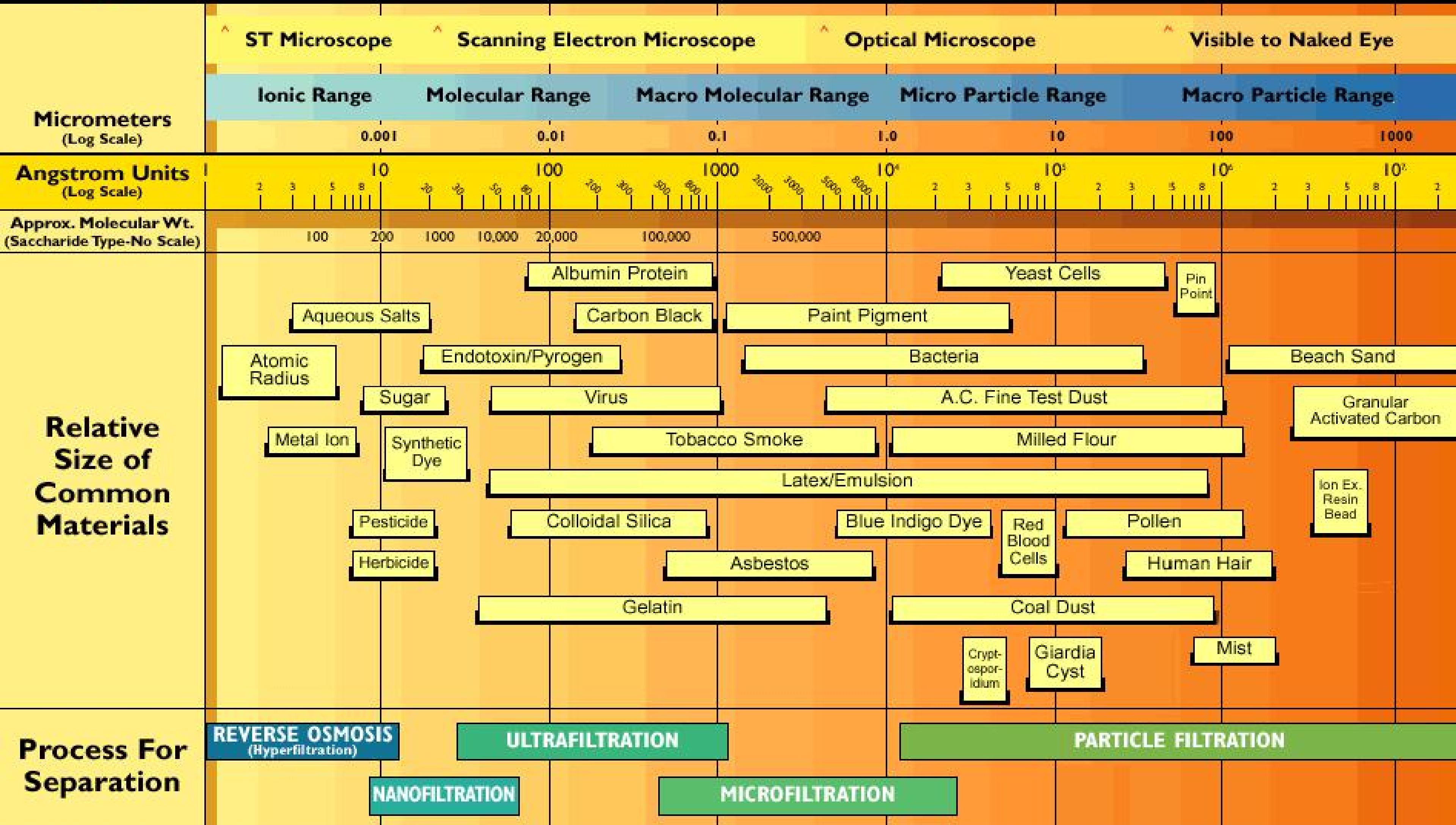


0.2 micron (200 nm)
99.9999% bacteria & 99.9% viruses
Adenoviruses 70-90 nm
Herpes 120-200 nm
Retro 80-100 nm
Parvovirus 20-30 nm
Prions be as small as 3 nm



OSMONICS

The Filtration Spectrum



Note: 1 Micron (1×10^{-6} Meters) ≈ 4×10^{-5} Inches (0.00004 Inches)
1 Angstrom Unit = 10^{-10} Meters = 10^{-4} Micrometers (Microns)

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Nitrogen Separation Applications

Oil Well, Pressure Control

Inert Atmosphere

Military Fuel Tank Protection

Hydrogen Separation Applications

Hydrogen removal in refineries, ammonia plants, and olefin units.

CO_2 Separation Applications

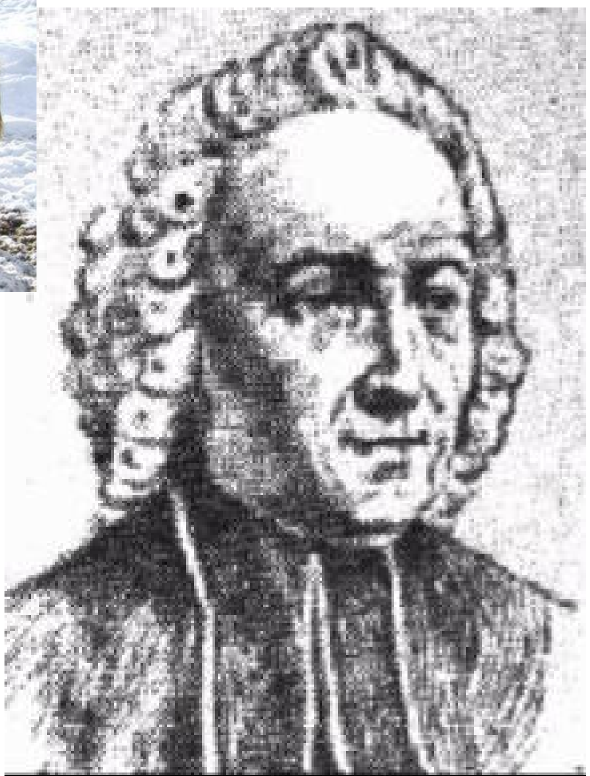
CO_2 removal in natural gas

CO_2 removal in synthetic natural gas

Advantages of Membranes

- **Energy Savings**
  - Transportation
  - Separations (20% less for EtOH)
  - Complement Existing Infrastructure
- **Flexibility-Modular Design**
  - Additional Units Easily Added
  - Easily Upgraded with New Technology

First Membrane



Jean Antoine Nollet
1700-1770

Another early Membrane: Sausage Casings

Submucosal collagen

Membranes

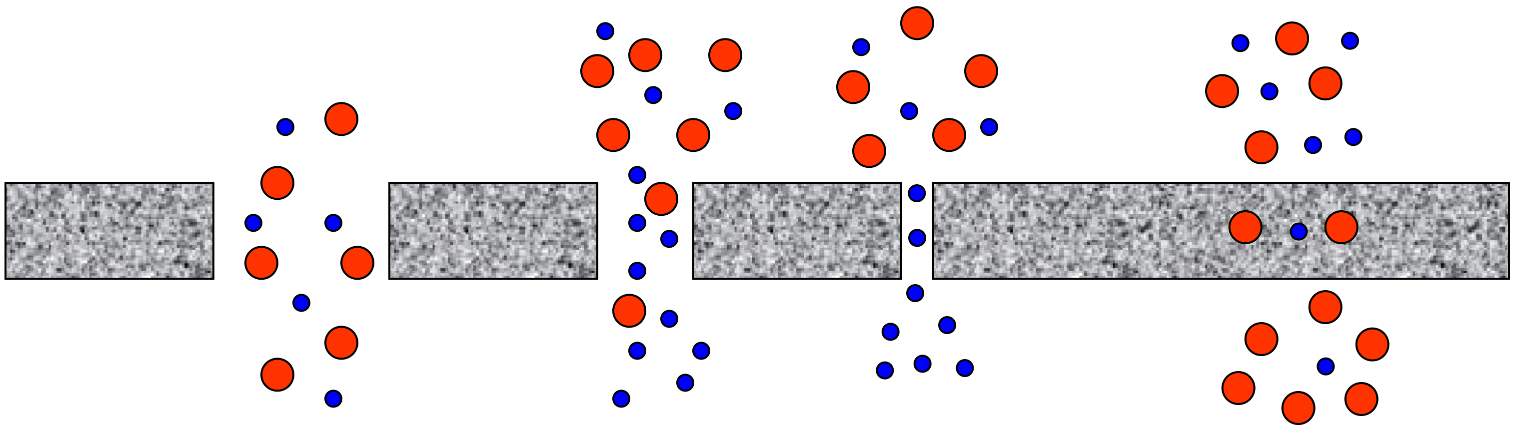
A thin barrier that permits selective mass transport

- Desalination
- Ultrafiltration
- Dialysis
- Sour gas and carbon dioxide removal
- Uranium isotope separation
- Oxygen-Nitrogen separation
- Pervaporation
- Batteries and fuel cells

Separation Mechanisms

Morphology Dictates Mode of Permeation and Separation

Pore Size: 0.1-10 m 0.002-0.1 m 0.0002-0.002 m Polymer Void Vol.



Convective Flow
No Separation

Knudsen Flow
Rate µ $(M_w)^{-1/2}$

Molecular Sieve
Rate µ Kinetic diameter

Dense Membrane
Solution Diffusion

Separation Mechanisms

Driving Forces:

- Concentration Gradient
- Pressure Gradient
- Electrical Potential Gradient

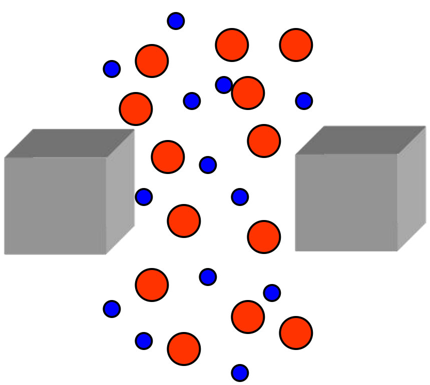
Separation:

- Differences in Solubility (dense membranes)
- Facilitated Transport (biological membranes)
- Size (molecular sieve membranes)
- Molecular Weight (porous membranes)
- Differences in Reactivity

Membrane Morphologies

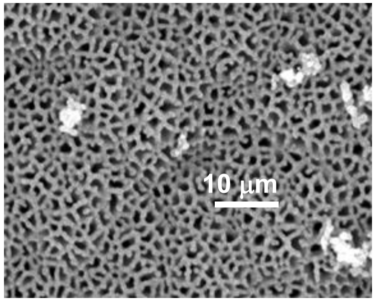
Pore Size:

1,000 – 100,000 Å

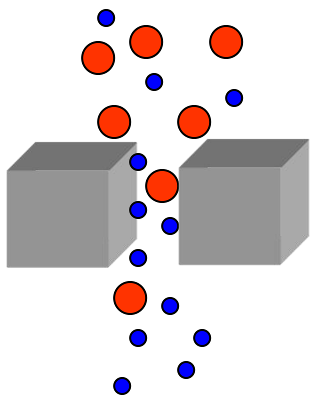


Poiseuille Flow

No Separation



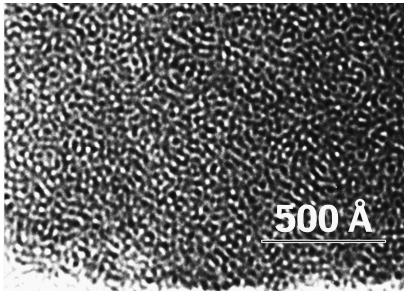
20 – 1,000 Å



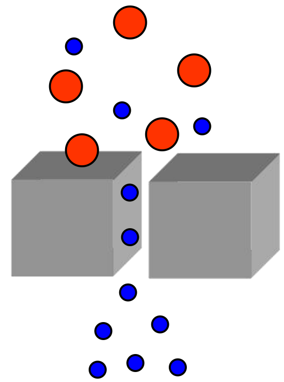
Knudsen Diffusion

$P \propto (MW)^{-1/2}$

$\alpha_{A/B} = (MW_A/MW_B)^{1/2}$



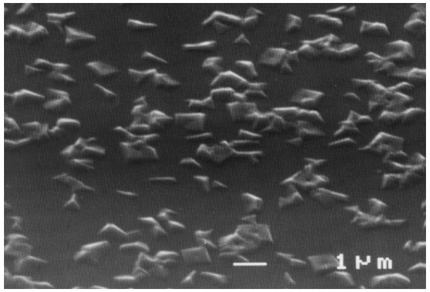
2 - 20 Å



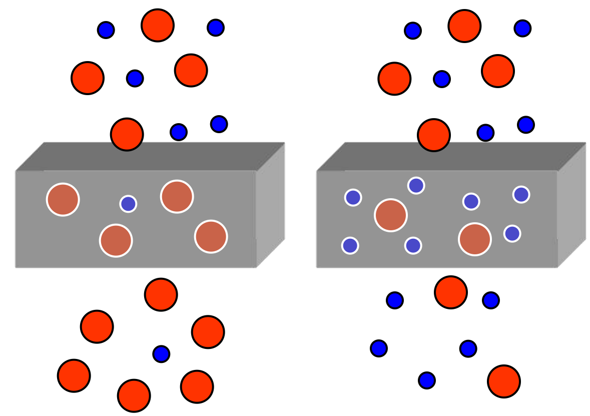
Molecular Sieve

$P \propto KD$

$\alpha_{A/B} = P_A/P_B$



Dense Membrane

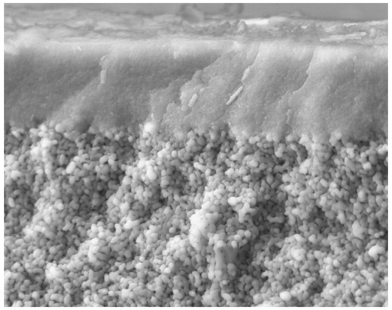


Rubbery **Glassy**

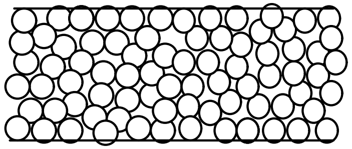
Solution Diffusion

$P = DS$

$\alpha_{A/B} = P_A/P_B$

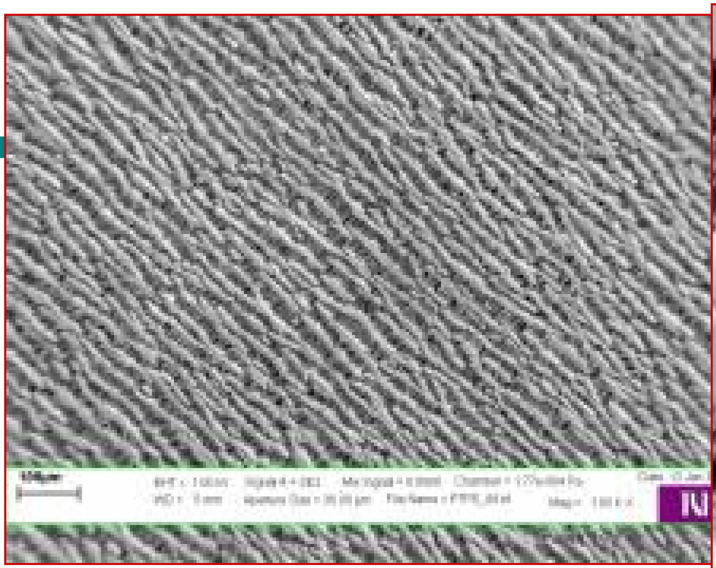


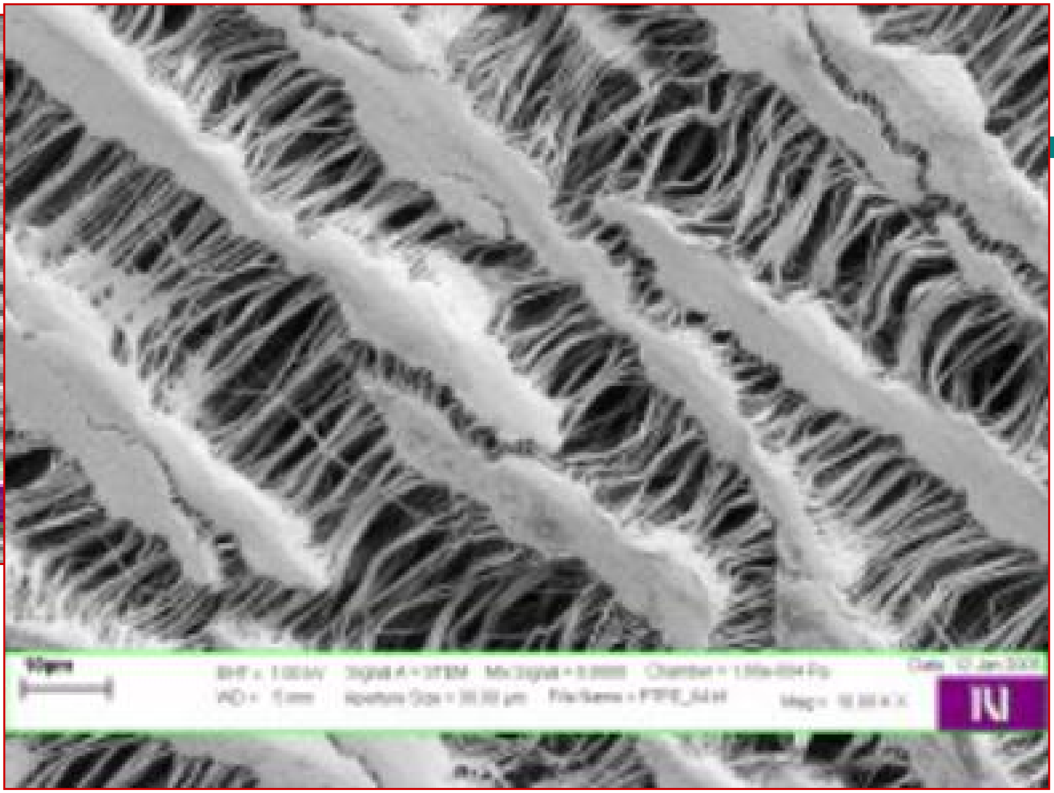
Preparation of Isotropic Microporous Membranes



Traditionally membranes with pores in the “micron” size range (0.1-10 microns); Really includes isotropic ceramic and metal Membranes.

- **Prepared by variety of methods**:
  - ⇒ Solution-Precipitation (Includes sol-gel)
  - ⇒ Irradiation/leaching of polymer films (Nucleopore)
  - ⇒ Expanded Films: Orientation/stretching crystalline polymers (Gore-Tex)
  - ⇒ Template Leaching (Microporous glass)
- Limitations:
  - ⇒ Large pores; not useful for gas or small molecule separations
- **Applications**:
  - ⇒ Microfiltration





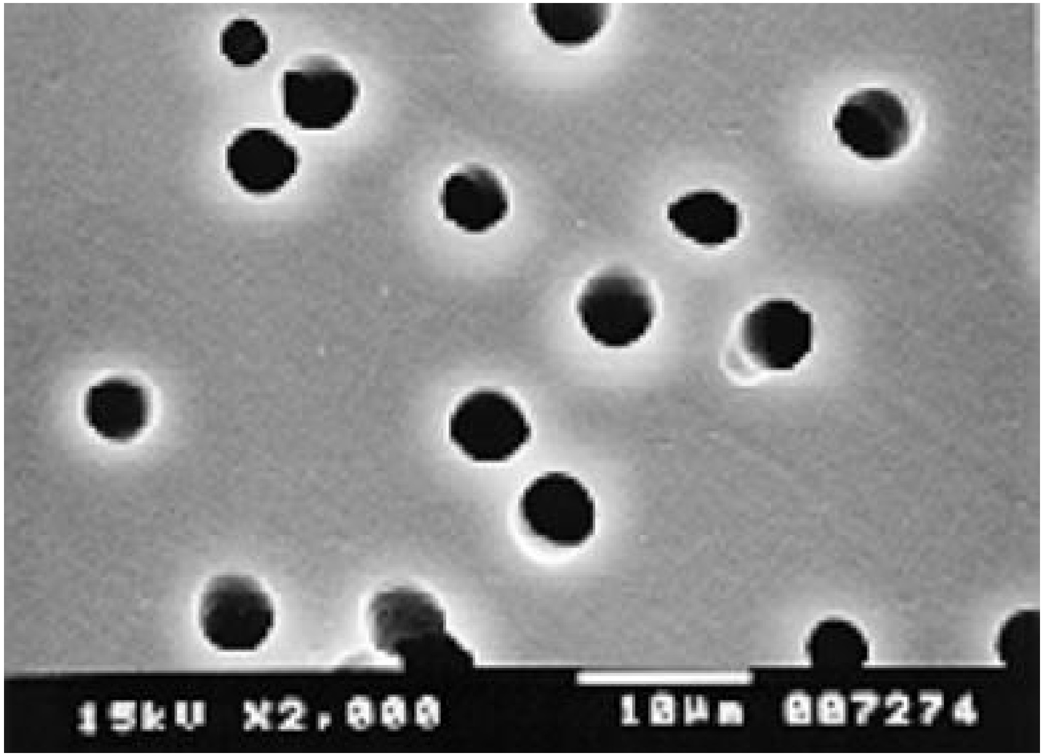
Teflon

(Membrane)

Pathogen Filters

Prepared by stretching the polymer film below T_m and above T_g

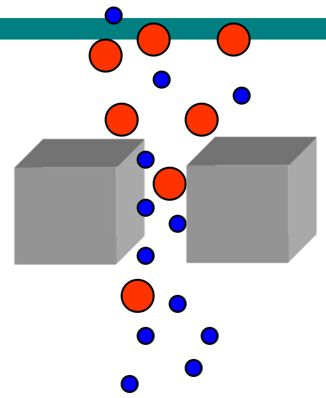
Ion Etch Polycarbonate



Irradiate polycarbonate film with ionizing particles
Etch out broken fragments with base

Knudsen Diffusion

20 – 1,000 Å

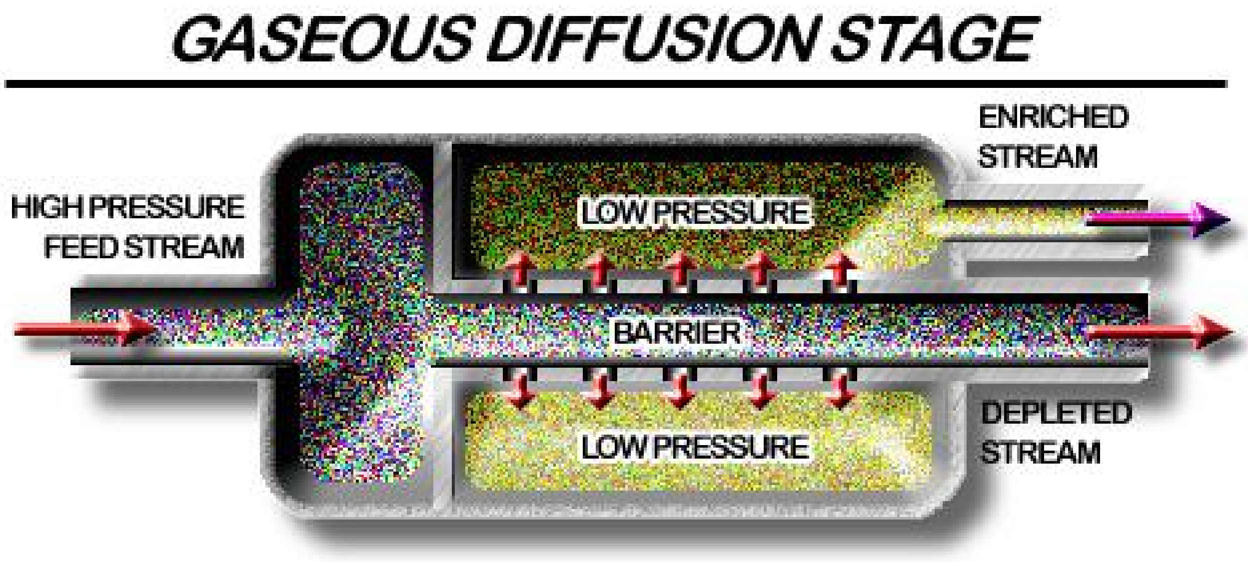


Knudsen Diffusion

$P \propto (MW)^{-1/2}$

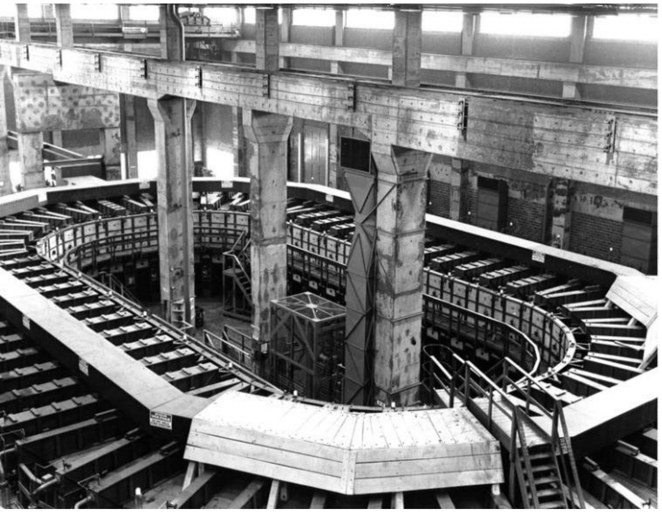
$\alpha_{A/B} = (MW_A/MW_B)^{1/2}$

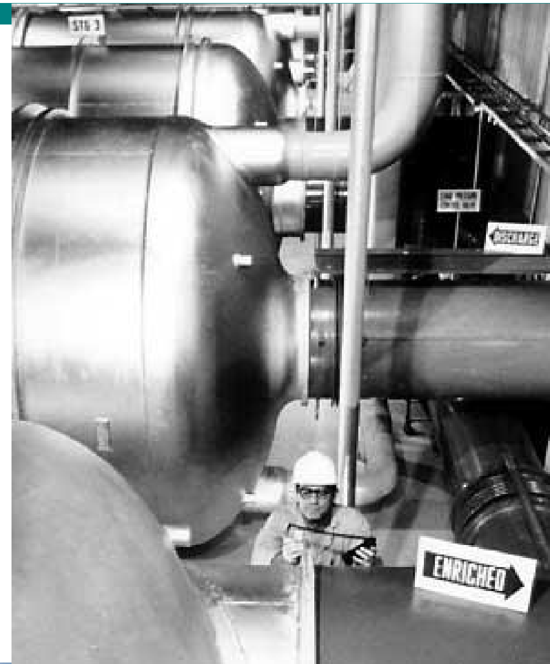
$^{238}UF_6$ from $^{235}UF_6$

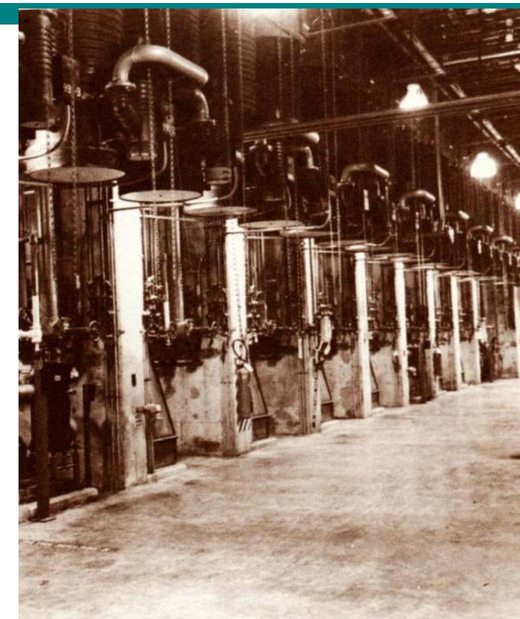


Knudsen Diffusion

$^{238}UF_6$ from $^{235}UF_6$





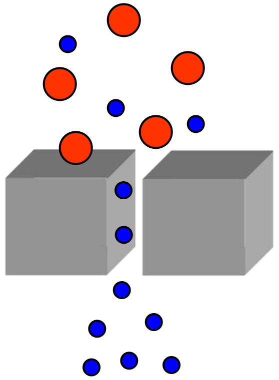


Porous Membranes: Molecular Sieving

Molecular Sieve

$P \propto KD$

$\alpha_{A/B} = P_A/P_B$



2 - 20 Å

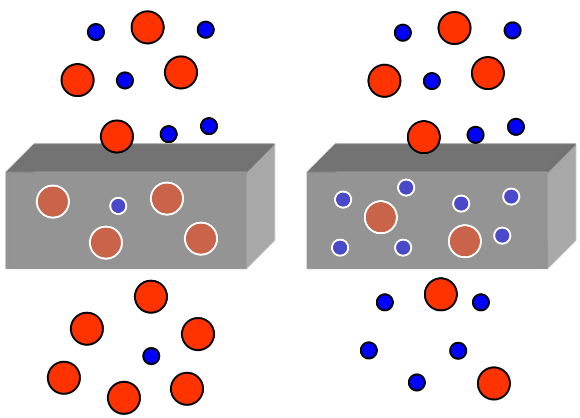
Molecular Sieve

Rate $\propto$ Kinetic diameter

	He	H_2	CO_2	CH_4	C_2H_4	C_3H_8	n-C_4H_{10}
Kinetic Diameter(nm):	0.26	0.289	0.33	0.38	0.39	0.43	0.43

Dense (non-porous) membranes

Dense Membrane



Rubbery **Glassy**

Solution Diffusion

$P = DS$

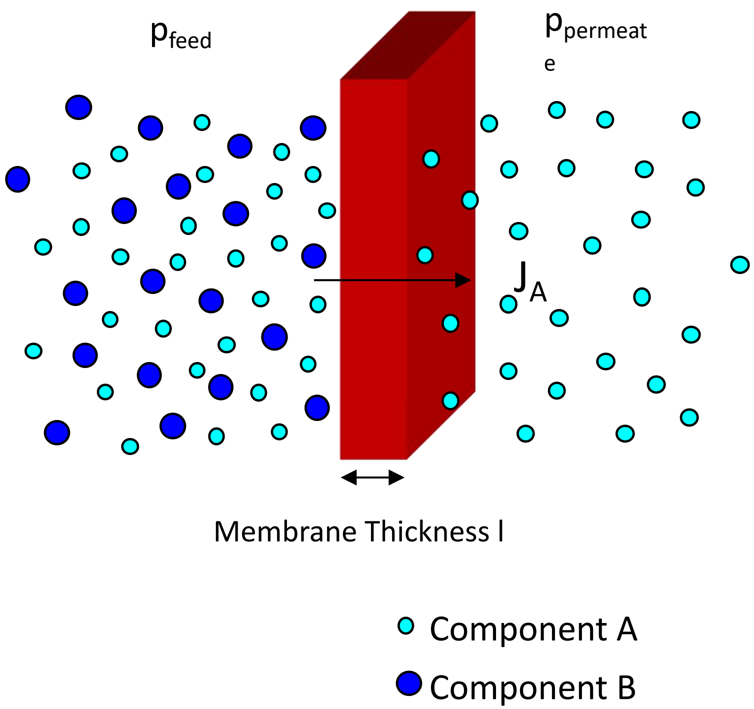
$\alpha_{A/B} = P_A/P_B$

Preparation of Dense Membranes

Most membranes are based on “dense” organic polymers

- **Free standing membranes:** > 20 microns thick, non-porous
  - $\Rightarrow$ Solution casting (10-20 wt%) with “hand casting knife”
  - $\Rightarrow$ Pressing between heated plates
- **Limitations**:
  - $\Rightarrow$ Poor flux-slow permeation
  - $\Rightarrow$ Intolerant of high pressure/temp
  - $\Rightarrow$ Chemical resistance can be an issue
- **Applications**:
  - $\Rightarrow$ Large scale use for packaging applications
  - $\Rightarrow$ Laboratory testing of new membrane materials

Non-Porous Membranes:



- Flux of A = $J_A = \dfrac{P_A\,[p_{feed}A - p_{perm}A]}{l}$

- Permeability of A = $P_A = D_A S_A$

  Where D_A = Diffusion Coefficient of A

  Where S_A = Solubility Coefficient of A

- Selectivity = $a_{A/B} = \left(\dfrac{D_A}{D_B}\right)\left(\dfrac{S_A}{S_B}\right)$

Mobility Selectivity

Solubility Selectivity

Sorption on upstream side
Diffusion down partial pressure gradient
Desorption on downstream side
Diffusion is rate determining step

Free Volume
T_g (Segmental Mobility)

Condensability

Nonporous (dense) membranes

- Predominant mechanism used in reverse osmosis (liquid feed), gas permeation (gas feed) and pervaporation (liquid and vapor feeds)
- Gas or liquid components absorb into the membrane at the upstream face, diffuse through the solid membrane, and desorb at the downstream face.
- $D_L < D_G$

  Diffusivities of solutes in solids < D of solutes in liquids.

  At 1 atm, 25°C diffusivities (cm^2/s):

  water vapor in air = 0.25

  water in ethanol (liquid) = 1.2×10^{-5}

  dissolved water in cellulose (solid) = 1.0×10^{-8}

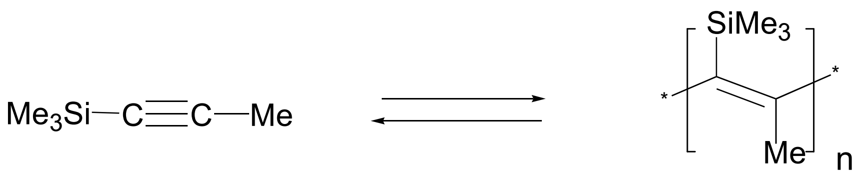
Gas Separations

Membrane Selectivity: $a_{ij} = \dfrac{P_i}{P_j} = \left(\dfrac{D_i}{D_j}\right)\left(\dfrac{k_i}{k_j}\right)$

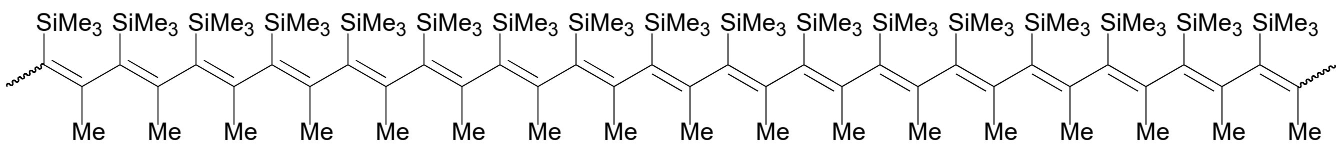
P = permeability
D = diffusion coefficient
k = sorption coefficient

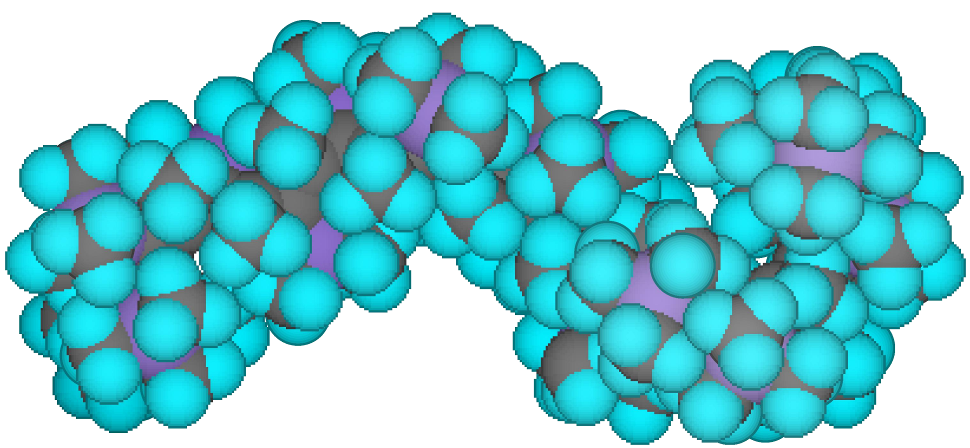
Membrane Material	P_{O_2}/P_{N_2}	P_{CO_2}/P_{CH_4}
Silicone Rubber	**2.0**	**3.38**
Natural Rubber	**2.96**	**4.4**
Butyl Rubber	**3.96**	**6.6**
Polycarbonate	**5.30**	**24.0**
Cellulose Acetate	**3.5**	**31.0**

PolyPTMSP: Clues to Engineering Void Volume



Polytrimethylsilylpropyne

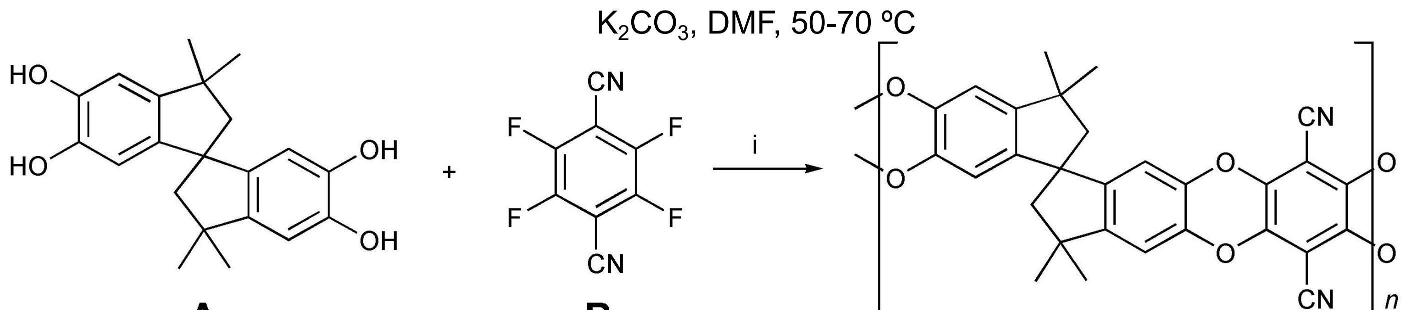




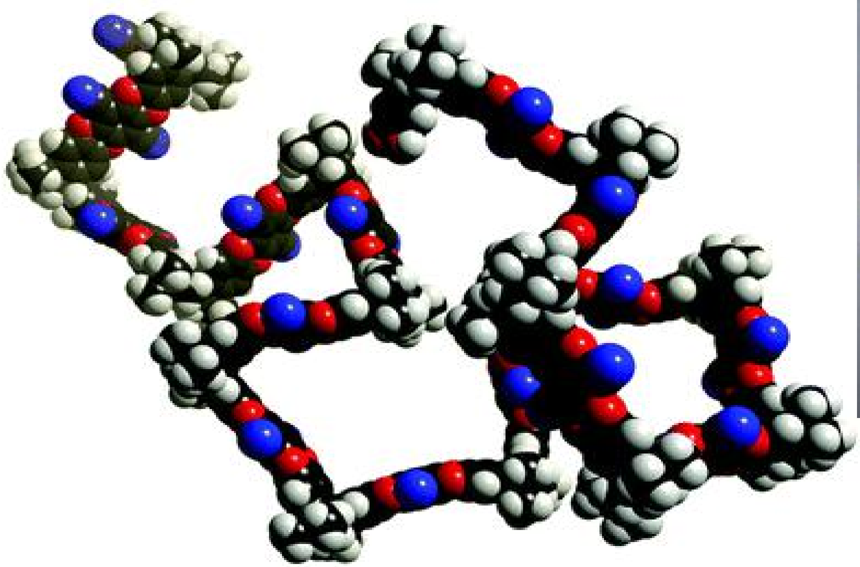
Non-Conjugated
High FFV (0.29-0.34 mL/mL)
Glassy Polymer (T_g >250° C)
Loss with Relaxation

Highest Gas Permeabilty for Any Polymer
21,000 Barrers O_2 ; 14,800 Barrers N_2; 12,700 Barrers He

Polymers with Inherent Microporosity (PIM’s)



A B PIM -1

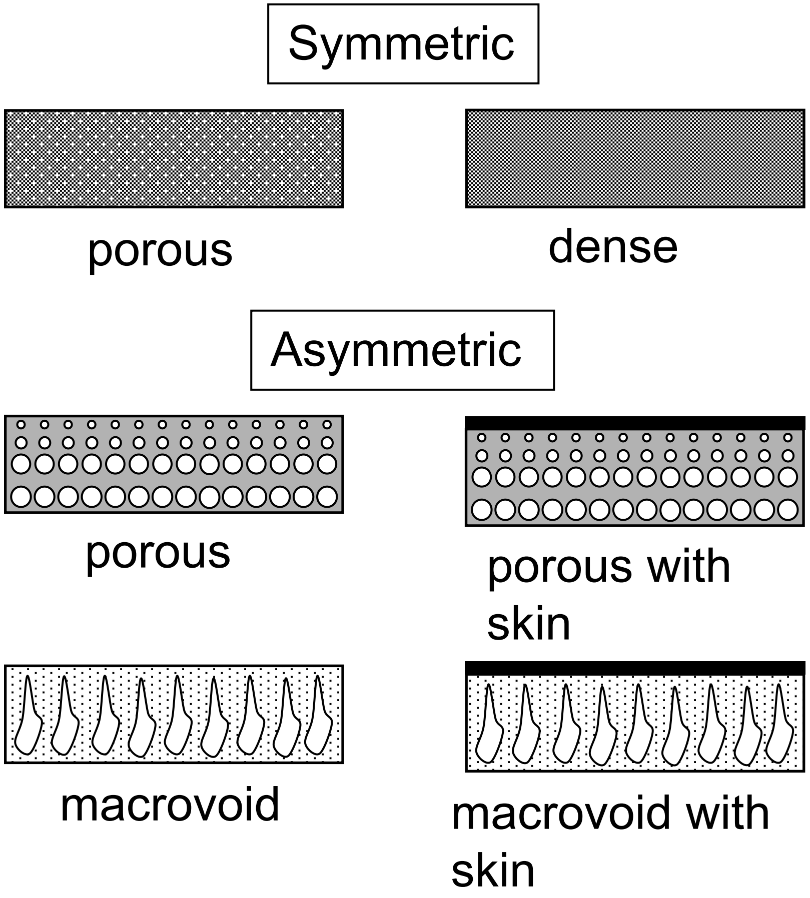


Commercial Gas Separations

H_2/N_2 Ammonia Purge Gas	Polysulfone Hollow Fibers	Monsanto (Permea)
CO_2/CH_4	Cellulose Acetate	Cynara, Separex
H_2/CH_4	Polyaramide (α = 1500)	DuPont
N_2/Air	Polyolefin Hollow Fiber Polysulfone " " Polyphenyleneoxide fibers	Dow Permea Delair
H_2S/CH_4	Cellulose Acetate	Cynara, Separex
He/CH_4		
Hydrocarbons/air		
CO/H_2	Polysulfone Hollow Fibers Polyamide	 Ube

• >3500 million cubic feet/day • 66% is H_2 separation • 7% N_2/Air •

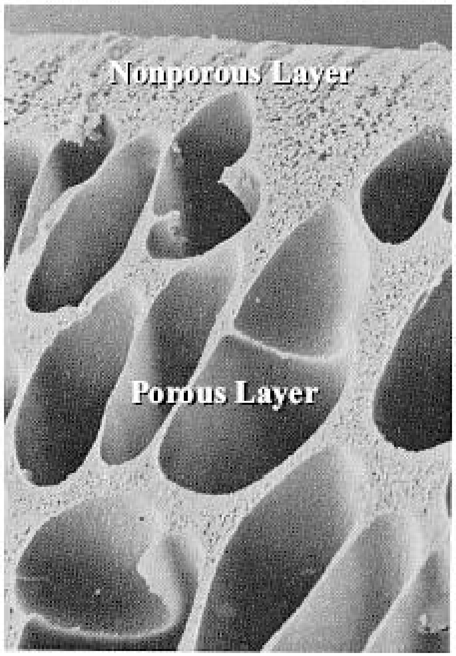
Thinner membranes means faster diffusion



Typical Membrane Structures
(Gas Separation)

Asymmetric membranes:
- very thin non-porous layer - selective
- thick, highly porous layer - mechanical support

Composite membranes:
- thin selective layer of one type polymer
- mounted on asymmetric membrane - support



Nonporous layer (selectivity)

Porous layer (stability)

Asymmetric membrane structure
(one type of material)

Selective layer

Asymmetric membrane

Composite membrane structure
(two types of materials)

Preparation of Asymmetric Membranes:

Revolution in Membrane Prep. And Performance: Loeb-Sourirajan Technique (In *Saline Water Conversion II*, D. R. Lloyd, Ed. ACS Symp. Series. 269; American Chemical Society: Washington, DC, 1963; p117.

- **Prepared by variety of methods**:

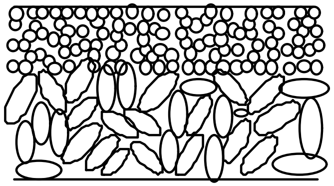
  ⇒ Solution-Precipitation or Phase Inversion

  » Thermal gelation

  » Solvent/Non-solvent Evaporation

  » Imbibition of Water Vapor

  » Non-Solvent Immersion

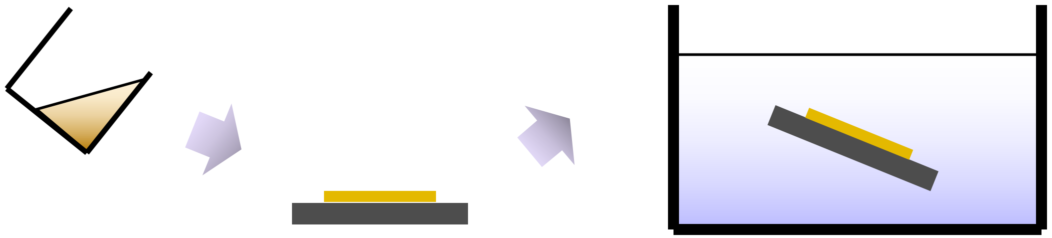


- **Applications**:

  ⇒ High flux asymmetric membranes have replace symmetric membranes in all applications except for microfiltration.

Asymmetric membrane formation

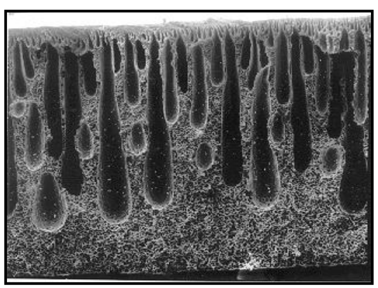
Phase separation by immersion precipitation



Polymer + solvent(s)

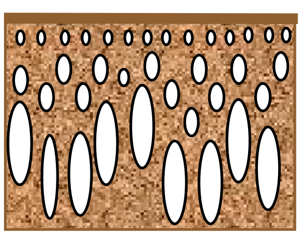
Nonsolvent (bath)

- Typical morphology of asymmetric membranes

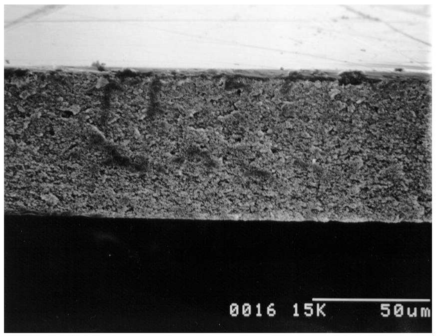


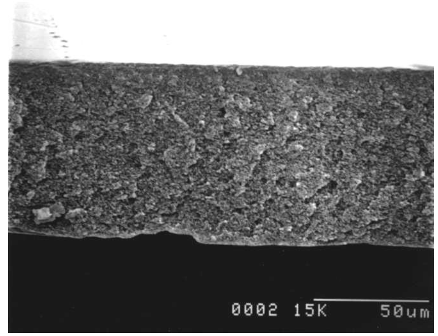
finger-like

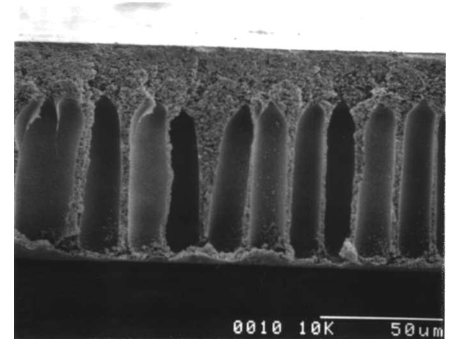
sponge-like



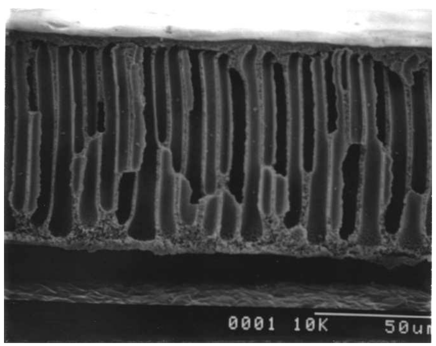
Membrane Morphology: cosolvent effect

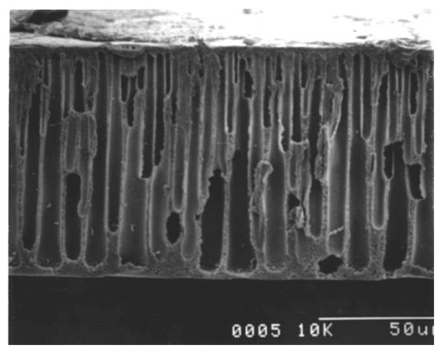


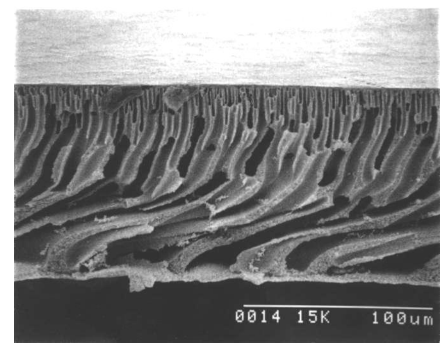




GBL/NMP: 100/0 80/20 60/40

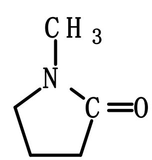




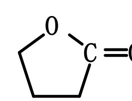


40/60 20/80 0/100

- Sponge-like → finger-like as NMP content increases
- Heat of mixing effect is clear

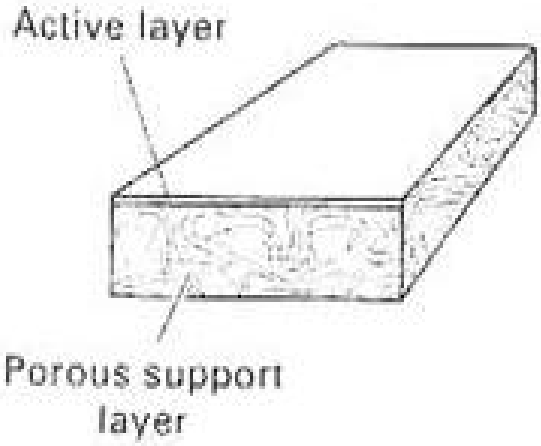


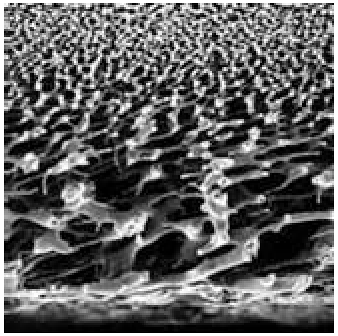
1-methyl-2-pyrrolidinone (NMP)



γ-butyrolactone (GBL)

Flat Sheet Membranes

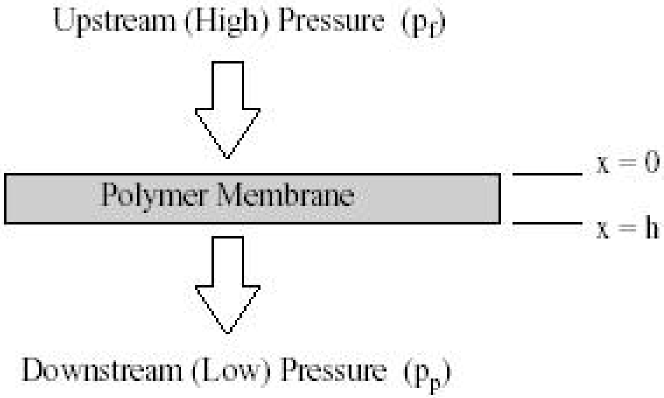




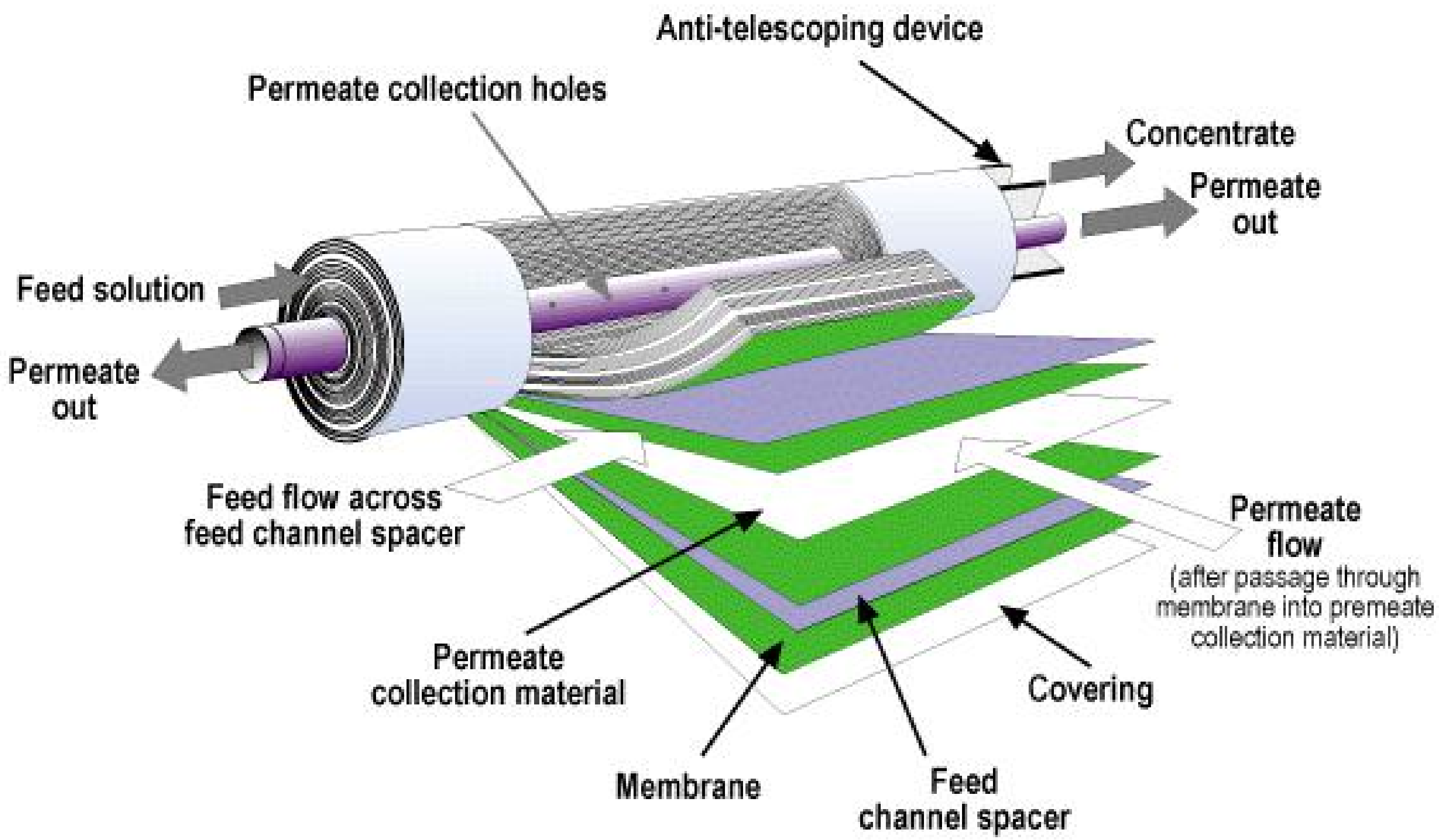
Asymmetric membranes

Used in laboratories for research

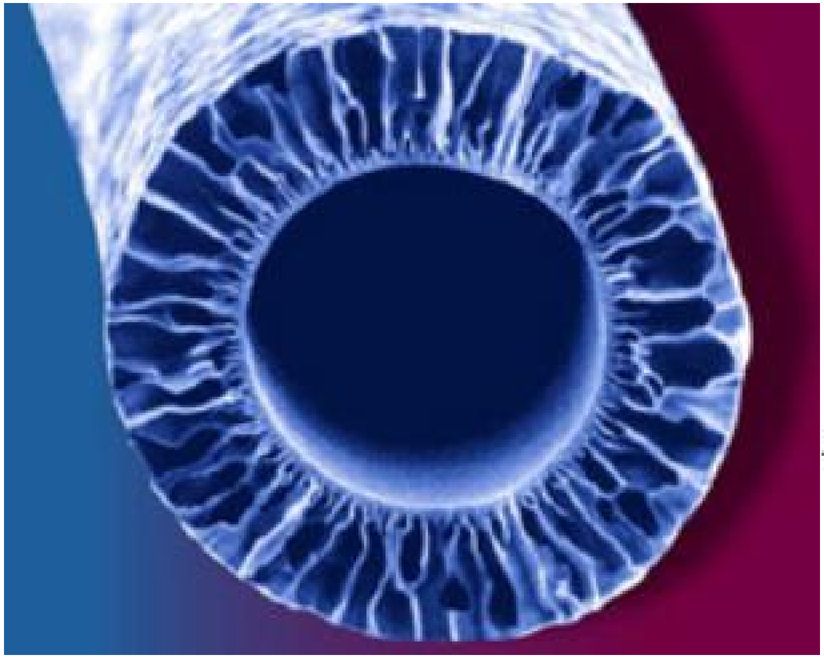
Easy to produce

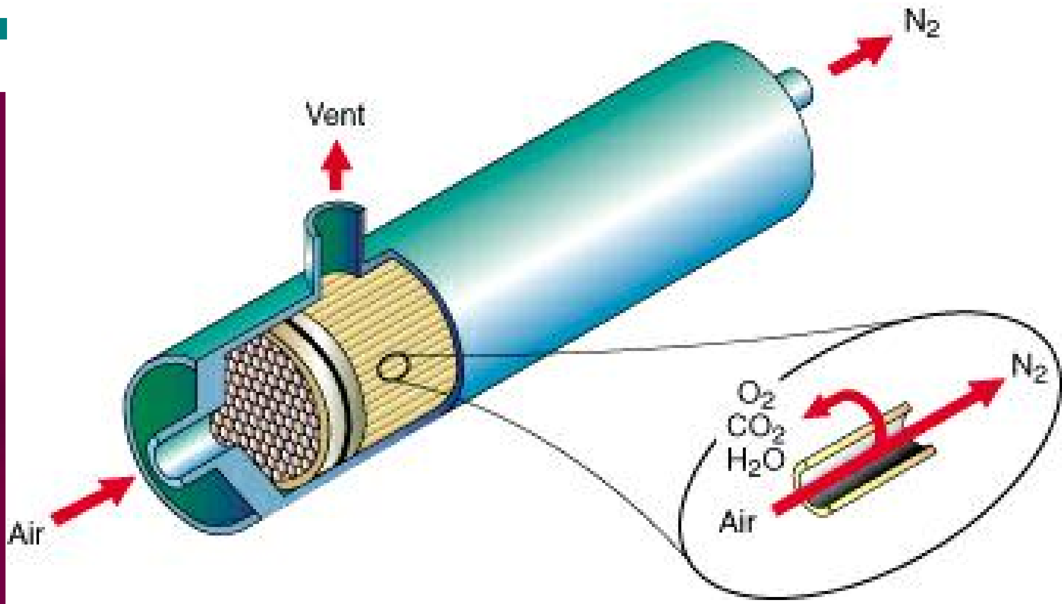


Membrane Technology



Hollow Fiber Membranes



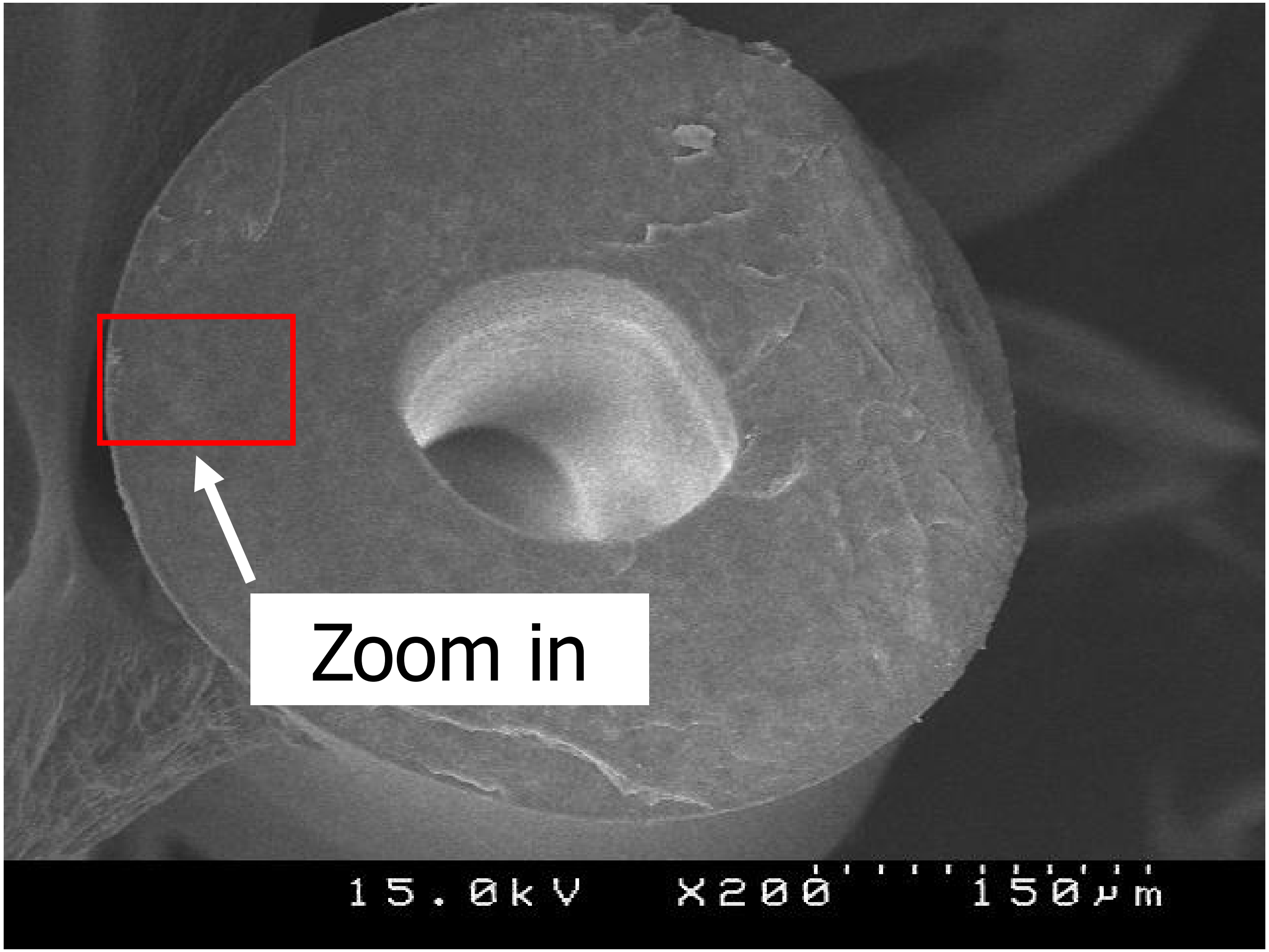


This are commonly used in industries

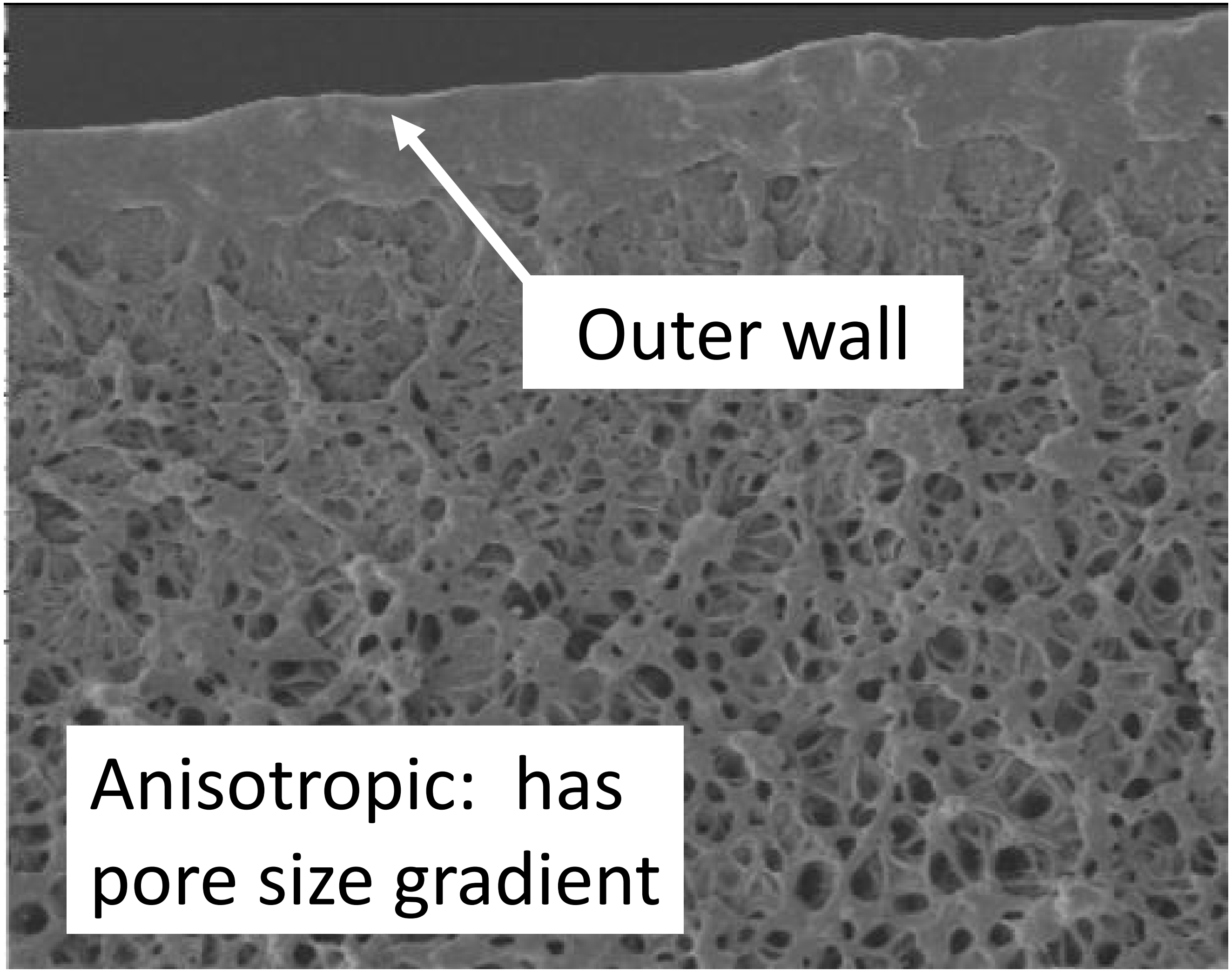
-High surface area

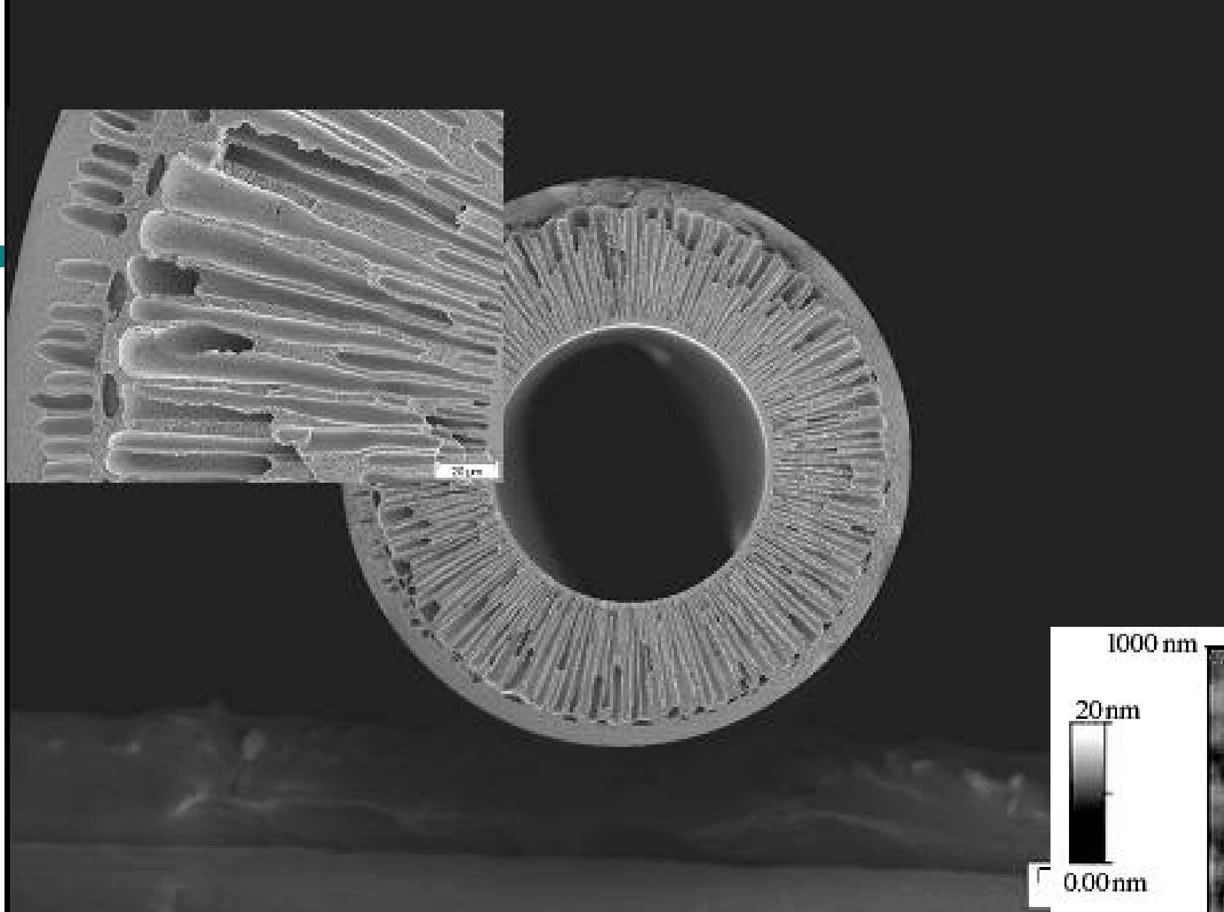
-Compactness

Hollow Fiber Membrane Cross-section



Hollow Fiber Membrane Microstructure





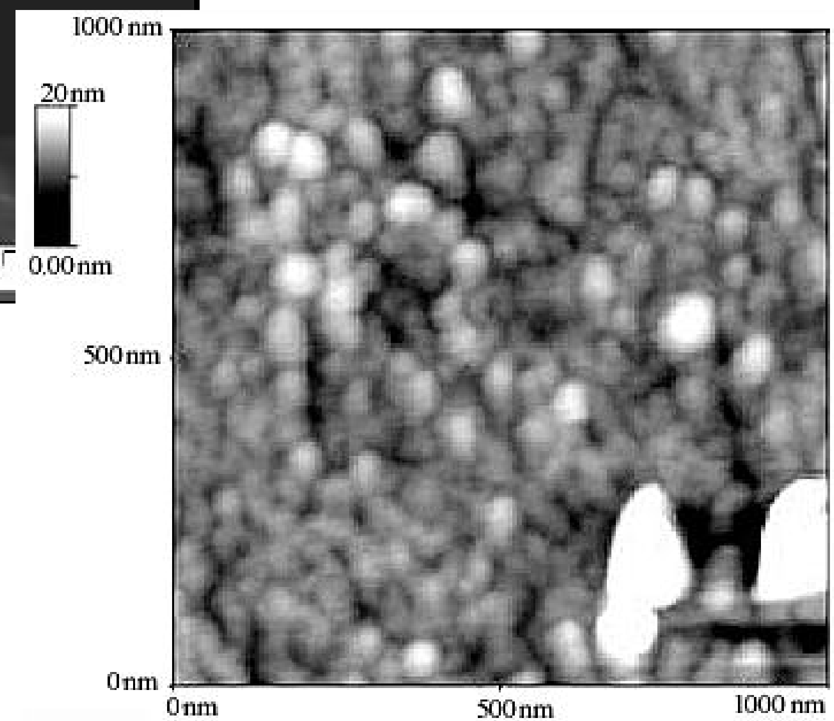
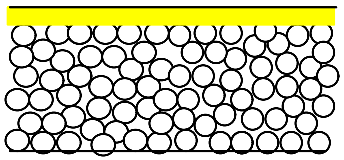


Figure 2. AFM image of the top surface of the PSf asymmetric ultrafiltration membrane (substrate). R_a = 1.6 nm.

Preparation of Interfacial Composite Membranes

Combination of Asymmetric and Dense Membranes

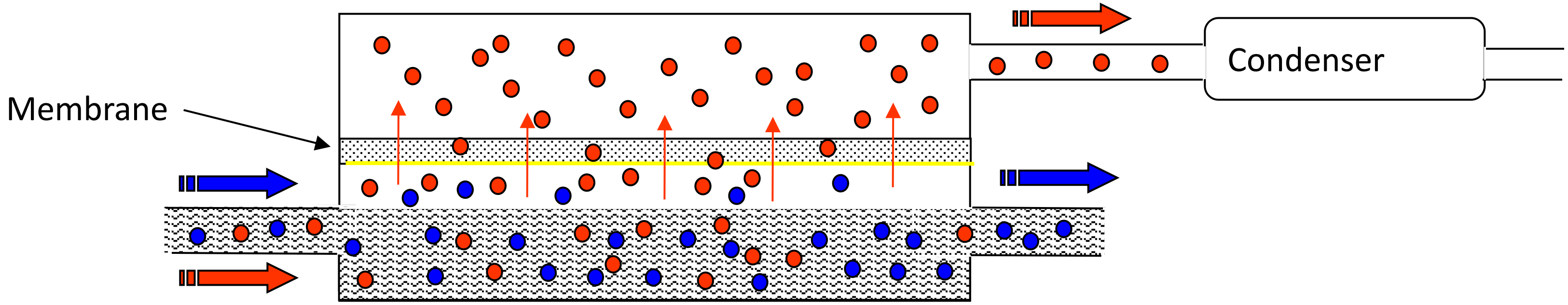
- Prepared by variety of methods:
  - $\Rightarrow$ Interfacial Polymerization
  - $\Rightarrow$ Solution Cast Composite Membranes
  - $\Rightarrow$ Plasma Polymerization (onto support)
  - $\Rightarrow$ Reactive Surface Treatment
  - $\Rightarrow$ Molecular leaching (ultra-microporous)
  - $\Rightarrow$ Sol-gel depositions



- Applications:
  - $\Rightarrow$ Gas separations, pervaporation, dialysis, etc..

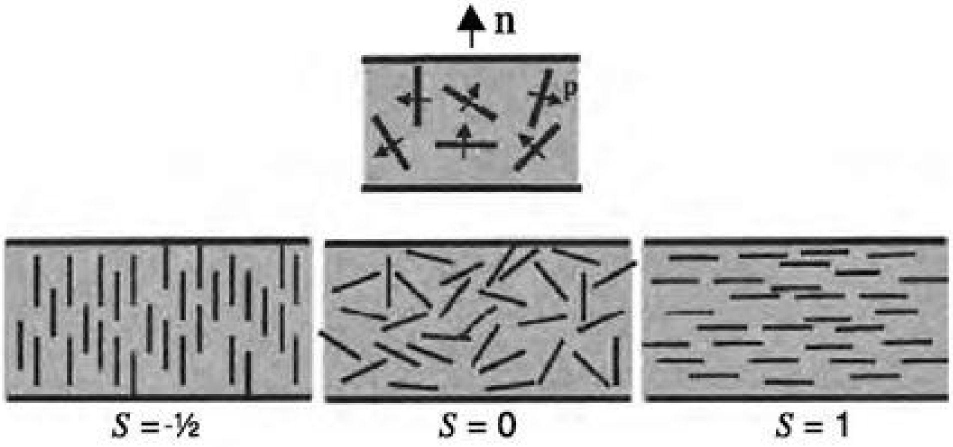
Pervaporation Membranes

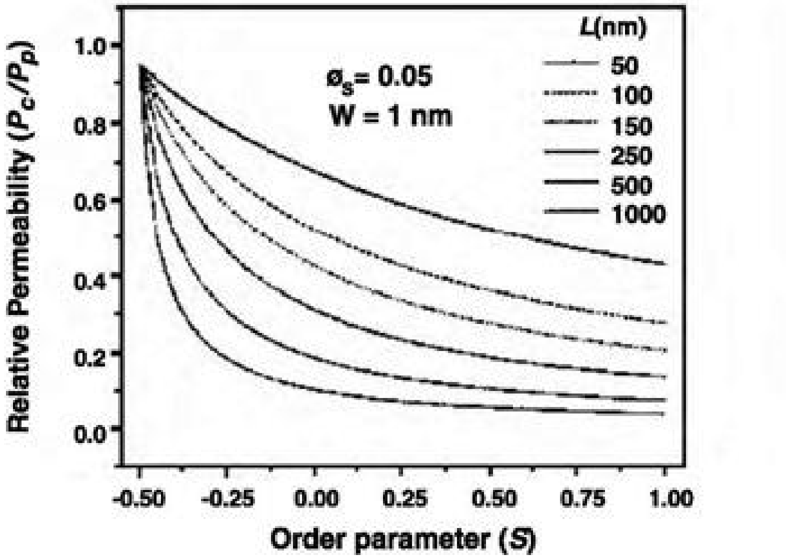
Membranes preferentially pass one or more components of a multicomponent liquid stream.



• Pressure gradient driven • Vapor phase transport • H_2O/EtOH •
• MeOH/MTBE-Isobutylene • Under develop. Aromatic/Hydrocarbon •

Polymer Nanocomposite Membranes





Research Issues with Membranes

- Selectivity
- Flux (Permeation Rate)
- Chemical Resistance
- Thermal Stability
- Cost-Materials and Replacement
- Fouling/Regeneration

Classification of synthetic membranes

Geometry	Bulk structure	Production method	Separation regime	Application
Hollow fiber Flat sheet	Symmetric	Stretched films Nucleation tracks Phase inversion	Porous	Dialysis ultra filtration Micro filtration
		Extrusion Solution casting	Non porous	Gas separation
	Asymmetric	Phase inversion	Porous	Dialysis ultra filtration Micro filtration
		Solution casting Interfacial polymerization Plasma polymerization	Non porous	Reverse osmosis Pervaporization Gas separation

HOMEWORK

- Introduce at least 2 polymer membranes and explain their applications
- Encourage you to give a short talk on what you find during next class.